

Task#3 Problem:

- a) Create a program that allows the user to enter values for the width and length of a room's floor in feet. The program outputs the area of the floor in square feet.

Hint:

$$\text{Area (sq ft)} = \text{Length (ft)} * \text{Width (ft)}$$

- b) Modify the program that computes floor area to compute and output the number of 6-inch square tiles needed to tile the floor. Then write a C++ program for this task.

Hint:

- *To find the exact number of tiles needed, divide the room's total square footage by the square footage of a single tile*
- *square footage of a single tile = Length of tile (ft) * Width of tile (ft)*
- *convert inch to feet = Divide the value by 12*

<u>Input</u>	<u>Processing</u>	<u>Output</u>
Width of the room in feet, length of the room in feet	Processing items: Area = Width*Length Tile Area = $(6.0/12)*(6.0/12)$ Number of Tiles=Area/Tile Area <u>Algorithm:</u> Step 1: Start Step 2: Enter the width of the room in feet, enter the length of the room in feet Step 3: Calculate Area = Width*Length Step 4: Calculate Tile Area = $(6.0/12)*(6.0/12)$ Step 5: Calculate Number of Tiles=Area/Tile Area Step 6: Print products Step 7: Stop	The area of the room, the number of tiles needed.

Flowchart

